

Customer Health Scoring Model & Automation Spec

SECTION 08-customer-success TYPE document STATUS **active** OWNER Founder

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Purpose

This document defines the mathematical scoring engine and technical automation spec for Fly GACA Customer Health Scoring. It measures the health of B2B flight schools / Approved Training Organizations (ATOs), charter operators, MROs, and commercial operators licensing Fly GACA Pro for cadet cohorts.

The composite score is a single 0–100 metric calculated automatically every week, driving proactive Customer Success triggers (onboarding, at-risk intervention, renewal, expansion).

It operationalizes [health-dashboard-spec.md](#) and [customer-health-dashboard.xlsx](#).

Read "Licensed Seats" as package capacity. Schools buy annual packages, not seats — Cohort (SAR 12,000/yr, up to 25 seats, one 90-day intake), Academy (SAR 39,000/yr, up to 100 seats), Institution (from SAR 72,000). Every seat ratio below is measured against the package's capacity ceiling (`orgs.seat_limit`), which is a contracted maximum the school pays for whether or not it fills. See [03-finance/monetization.md](#).

Composite Formula & Dimension Weights

The composite customer health score (H) is a weighted linear combination of five normalized dimension scores ($S_i \in [0, 100]$):

$$[H = 0.30 \cdot S_{\text{usage}} + 0.20 \cdot S_{\text{engagement}} + 0.20 \cdot S_{\text{support}} + 0.15 \cdot S_{\text{financial}} + 0.15 \cdot S_{\text{sentiment}}]$$

Dimension	Weight ((w _i))	Telemetry Source	Primary Indicator
Product Usage ((S _{usage})))	30%	Cloud SQL: <code>study_progress.summary</code> joined to the <code>org_seats</code> roster (the same query behind <code>GET /api/org/:orgId/cohort-readiness</code>)	Seat activation, mock exam volume, Captain Adel AI queries

Dimension	Weight ((w _i))	Telemetry Source	Primary Indicator
Engagement ((S _{{\text{engagement}}})))	20%	Web analytics & CS CRM logs	Instructor dashboard logins, session depth, QBR attendance
Support Health ((S _{{\text{support}}})))	20%	Support Ticketing API (Zendesk)	Ticket volume, resolution time, SLA breaches, escalation rate
Financial / Contract ((S _{{\text{financial}}})))	15%	Accounting DB / ZATCA Billing + orgs.seat_limit	Package capacity fill, invoice payment timeliness
Sentiment ((S _{{\text{sentiment}}})))	15%	In-app NPS / CSAT & QBR surveys	Cadet NPS, instructor CSAT, executive champion strength

5-Factor Mathematical Calculations

1. Product Usage Score ((S_{{\text{usage}}})) — Weight: 30%

Evaluates active participation and learning velocity across the cadet roster over a rolling 7-day window.

$$[S_{\text{usage}} = 0.40 \cdot U_{\text{active}} + 0.30 \cdot U_{\text{exams}} + 0.30 \cdot U_{\text{ai}}]$$

Where:

- **Active Seat Ratio ((U_{{\text{active}}}))**: measured against **assigned roster seats**, not package capacity — a Cohort school that provisions 18 of its 25 seats should not be scored as 28% idle for capacity it deliberately left unused. Capacity fill is a separate, financial signal (§4).

$$[U_{\text{active}} = \min\left(100, \frac{\text{Active Cadets (7d)}}{\text{Assigned Roster Seats}} \times 100\right)]$$
- **Mock Exam Completion Velocity ((U_{{\text{exams}}}))**:

$$[U_{\text{exams}} = \min\left(100, \frac{\text{Completed Mock Exams (7d)}}{\text{Assigned Roster Seats} \times 1.5} \times 100\right)]$$
- **AI Instructor Engagement ((U_{{\text{ai}}}))**:

$$[U_{\text{ai}} = \min\left(100, \frac{\text{Captain Adel Queries (7d)}}{\text{Assigned Roster Seats} \times 4.0} \times 100\right)]$$

Granularity warning on Cohort accounts. With a 25-seat ceiling, one cadet is worth 4 percentage points of any seat ratio, and a full Cohort roster of 25 means an 85% target is unreachable except at 22 seats (88%) — 21 seats is 84% and misses. Thresholds expressed as percentages therefore do not mean what they mean on a 100-seat Academy account.

State Cohort triggers in cadets, not percent (see the alert matrix below), and never open an at-risk case on a one-seat move.

2. Engagement Score (($S_{\text{engagement}}$)) — Weight: 20%

Measures organizational alignment, instructor governance, and executive touchpoints over a rolling 30-day window.

$$[S_{\text{engagement}} = 0.40 \cdot E_{\text{instructor}} + 0.30 \cdot E_{\text{depth}} + 0.30 \cdot E_{\text{qbr}}]$$

Where:

- **Instructor Governance (($E_{\text{instructor}}$)):** $[E_{\text{instructor}} = \min\left(100, \frac{\text{Instructor Dashboard Logins (30d)}}{\text{Provisioned Instructors}} \times 4.0\right) \times 100]$
- **Cadet Study Session Depth ((E_{depth})):** $[E_{\text{depth}} = \min\left(100, \frac{\text{Average Session Duration (mins)}}{25.0} \times 100\right)]$
- **Executive QBR Participation ((E_{qbr})):** $[E_{\text{qbr}} = \begin{cases} 100 & \text{Head of Training / Sponsor attended scheduled QBR} \\ 60 & \text{Designated CFI / Admin attended QBR} \\ 0 & \text{QBR missed, rescheduled >2x, or declined} \end{cases}]$

3. Support Health Score ((S_{support})) — Weight: 20%

Quantifies technical friction and customer support load over a rolling 30-day window.

$$[S_{\text{support}} = \max\left(0, 100 - \left(25 \cdot T_{\text{P1}} + 10 \cdot T_{\text{P2}} + 5 \cdot T_{\text{P3}} + 15 \cdot S_{\text{breach}}\right)\right)]$$

Where:

- (T_{P1}): Unresolved P1 Critical tickets (system outage / auth blocking) in last 30d.
- (T_{P2}): Unresolved P2 High tickets (roster sync / score saving error) in last 30d.
- (T_{P3}): Open P3 Medium/Low tickets in last 30d.
- (S_{breach}): First-response or resolution SLA breaches in last 30d.

4. Financial / Contract Health Score (($S_{\text{financial}}$)) — Weight: 15%

Tracks package capacity fill and invoice payment compliance.

$$[S_{\text{financial}} = 0.50 \cdot F_{\text{fill}} + 0.50 \cdot F_{\text{payment}}]$$

Where:

- **Package Capacity Fill ((F_{\text{fill}}))**: $[F_{\text{fill}} = \min\left(100, \frac{\text{Assigned Roster Seats}}{\text{Package Capacity (} \text{orgs.seat_limit} \text{)}} \times 100 \right)]$ Under per-seat licensing an under-filled contract meant the school was paying for air, and a low ratio was a straightforward renewal risk. Under package pricing the school pays SAR 12,000 either way, so this metric no longer measures overspend — it measures **how much of the contracted value the school is actually taking**, and it reads in two directions:
 - **Low fill** ($\ll$ capacity) is a *value-realization* risk, not a billing one — the school is leaving capacity on the table and will feel the renewal is expensive. Route it to the onboarding/at-risk motion, not to a discount conversation.
 - **Fill at or near capacity** is an **upgrade signal**, since the ceiling is hard: the provisioning API refuses any call that would push the roster past `seat_limit`, so a school at 25 of 25 physically cannot enrol its next cadet without moving to Academy.
- **Payment Compliance ((F_{\text{payment}}))**: $[F_{\text{payment}} = \max\left(0, 100 - \max\left(0, \text{Days Past Due} - 14\right) \times 5\right)]$ (100 pts for payment on time or within 14-day grace period; deducts 5 pts per day thereafter).

5. Sentiment Score ((S_{\text{sentiment}})) — Weight: 15%

Captures subjective customer satisfaction, NPS surveys, and executive relationship stability.

$$[S_{\text{sentiment}} = 0.50 \cdot N_{\text{nps}} + 0.30 \cdot C_{\text{csat}} + 0.20 \cdot K_{\text{champion}}]$$

Where:

- **Net Promoter Score Normalization ((N_{\text{nps}}))**: $[N_{\text{nps}} = \begin{cases} 100 & \text{Promoter (Rating 9–10)} \\ 60 & \text{Passive (Rating 7–8)} \\ 0 & \text{Detractor (Rating 0–6)} \end{cases}]$
- **Customer Satisfaction Rating ((C_{\text{csat}}))**: Average CSAT rating (1–5 scale) normalized: $(C_{\text{csat}} = \text{CSAT Avg} \times 20)$.
- **Executive Champion Index ((K_{\text{champion}}))**: $[K_{\text{champion}} = \begin{cases} 100 & \text{Active executive champion \& public reference logo} \\ 70 & \text{Stable sponsor contact} \\ 30 & \text{Sponsor role changed / champion departed} \end{cases}]$

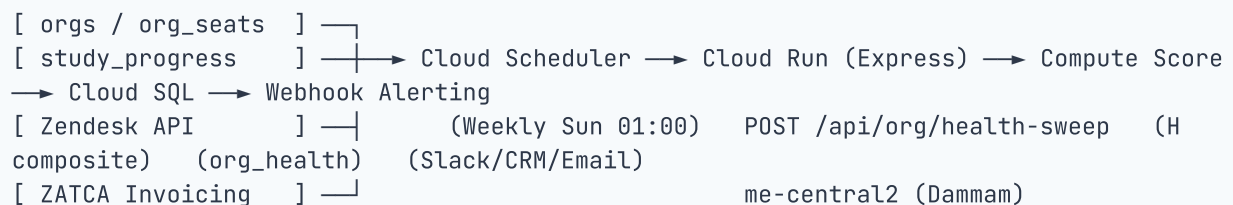
Score Bands & Recommended CS Playbooks

Band	Composite Score ((H))	Operational Meaning	Recommended Playbook	Automated System Trigger
Healthy	80–100	High adoption, strong ROI proof, zero SLA breaches	Expansion Playbook	Solicit testimonial / case study; notify AE on the band-upgrade trigger below

Band	Composite Score ((H))	Operational Meaning	Recommended Playbook	Automated System Trigger
Neutral	60–79	Stable account; minor usage drift or pending QBR	Renewal Playbook	Flag for CSM bi-weekly review; schedule mid-term review
At-Risk	0–59	Churn threat; idle roster or support friction	At-Risk Playbook	Immediate P1 CSM alert; initiate 14-day recovery plan

Technical Automation & Data Pipeline Spec

1. Architectural Overview



Everything on the left of that diagram except Zendesk is a table in the **Cloud SQL Postgres** instance behind the Cloud Run service — the same store the org dashboard reads. There is no separate telemetry database and no document store.

2. Scheduled Job Implementation Spec

- **Endpoint:** `POST /api/org/health-sweep` on the Cloud Run service, authenticated by the same `CRON_SECRET` header pattern the billing renewal job uses — **to be built**; the org router today exposes `/mine`, `/:orgId/cohort-readiness` and `/:orgId/provision-seats` only.
- **Schedule:** Cloud Scheduler, every Sunday at 01:00 UTC+3 (`cron: 0 1 * * 0`).
- **Execution Logic:**
 1. Select active orgs from `orgs` (with `seat_limit` = the purchased package capacity).
 2. Aggregate the roster: `org_seats` left-joined to `users`, `entitlements` and `study_progress.summary` over 7d/30d — seats exist before their invitee has an account, so every join is outer and the nulls are meaningful (an unclaimed seat is *provisioned but not active*, and must not be counted as an active cadet).
 3. Query Support API for ticket counts & SLA breaches in past 30d.
 4. Fetch invoice status from Billing DB (`daysPastDue`).
 5. Compute the 5 dimension scores (S_i) and composite score (H).
 6. Upsert the current row and append a dated snapshot row, both keyed by `org_id`.

3. Health Record Shape (one row per org, latest snapshot)

```
{
  "orgId": "3f0c...-uuid",
  "orgName": "oxford-saudia-dammam",
  "package": "cohort",
  "seatLimit": 25,
  "updatedAt": "2026-08-19T01:00:00Z",
  "compositeScore": 88.4,
  "band": "Healthy",
  "dimensions": {
    "usage": 88.0,
    "engagement": 82.5,
    "support": 100.0,
    "financial": 94.0,
    "sentiment": 76.0
  },
  "metrics": {
    "assignedSeats": 22,
    "activeCadets7d": 20,
    "activeSeatRatio": 0.91,
    "capacityFill": 0.88,
    "weeklyMockExamsPerCadet": 1.8,
    "monthlyInstructorLogins": 12,
    "openTicketsCount": 0,
    "daysPastDue": 0,
    "latestNpsScore": 9
  },
  "previousCompositeScore": 79.0,
  "delta7d": +9.4,
  "triggeredPlaybook": "expansion"
}
```

4. Automated Alert & Escalation Matrix

- **Condition 1 — Score Drops to At-Risk (($H < 60$)) or ($\Delta H \leq -15$) pts:**
 - System fires Slack alert to `#cs-at-risk-alerts` with detailed dimension breakdown.
 - Automatically creates Urgent P1 Task in Hubspot/Salesforce CRM assigned to CSM.
 - Triggers [At-Risk Playbook](#) workflow.
- **Condition 2 — Capacity Fill near the package ceiling & ($H \geq 80$):**
 - Trigger in **seats, not percent**, because the ceiling is small enough that a percentage is ambiguous: **Cohort — 22 or more of 25 assigned** (85% of 25 is 21.25, which no roster can be); **Academy — 85 or more of 100**; Institution — per contract.
 - System fires Slack alert to `#sales-expansion`.
 - Sends email to Account Executive with a **band-upgrade** recommendation — Cohort → Academy (SAR 12,000 → 39,000/yr), or Academy → Institution. There are no seat top-ups to sell: the only way past a package ceiling is the next package.
- **Condition 2b — Provisioning refused at the ceiling:**

- A `provision-seats` call rejected for exceeding `seat_limit` is the strongest expansion signal we get — the school tried to enrol a cadet and could not. Alert `#sales-expansion` the same day, independent of (H).
 - **Condition 3 — Support Health ($S_{\text{support}} < 50$):**
 - Sends automated high-priority ticket escalation to Head of Product & CS Lead.
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Related Documents

- [Health Dashboard Spec](#)
- [At-Risk Playbook](#)
- [Renewal Playbook](#)
- [Expansion Playbook](#)
- [Customer Success Overview](#)